Manuscript: Association between Janus kinase inhibitors therapy and mental health outcome in patients with rheumatoid arthritis: a systematic review and meta-analysis

SUPPLEMENTARY MATERIAL

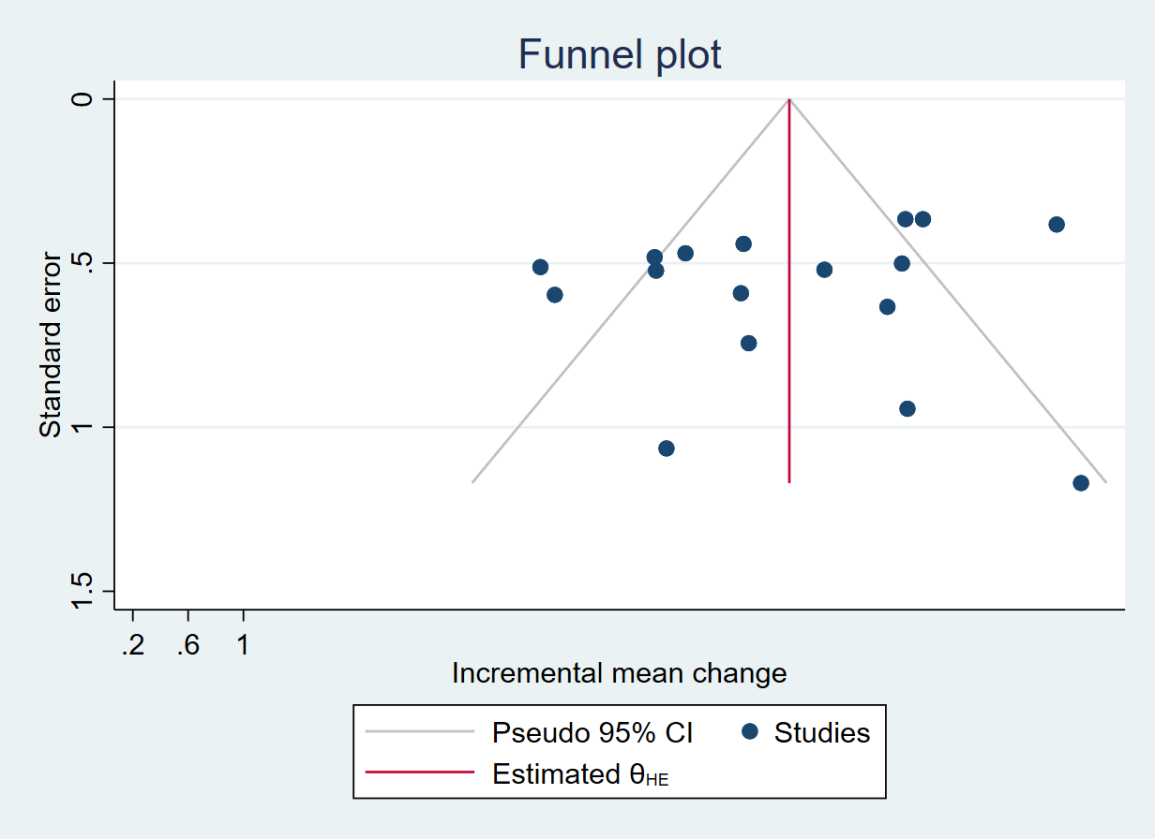
- Supplementary Figures
- Supplementary Tables
- PRISMA Checklist

SUPPLEMENTARY FIGURES:

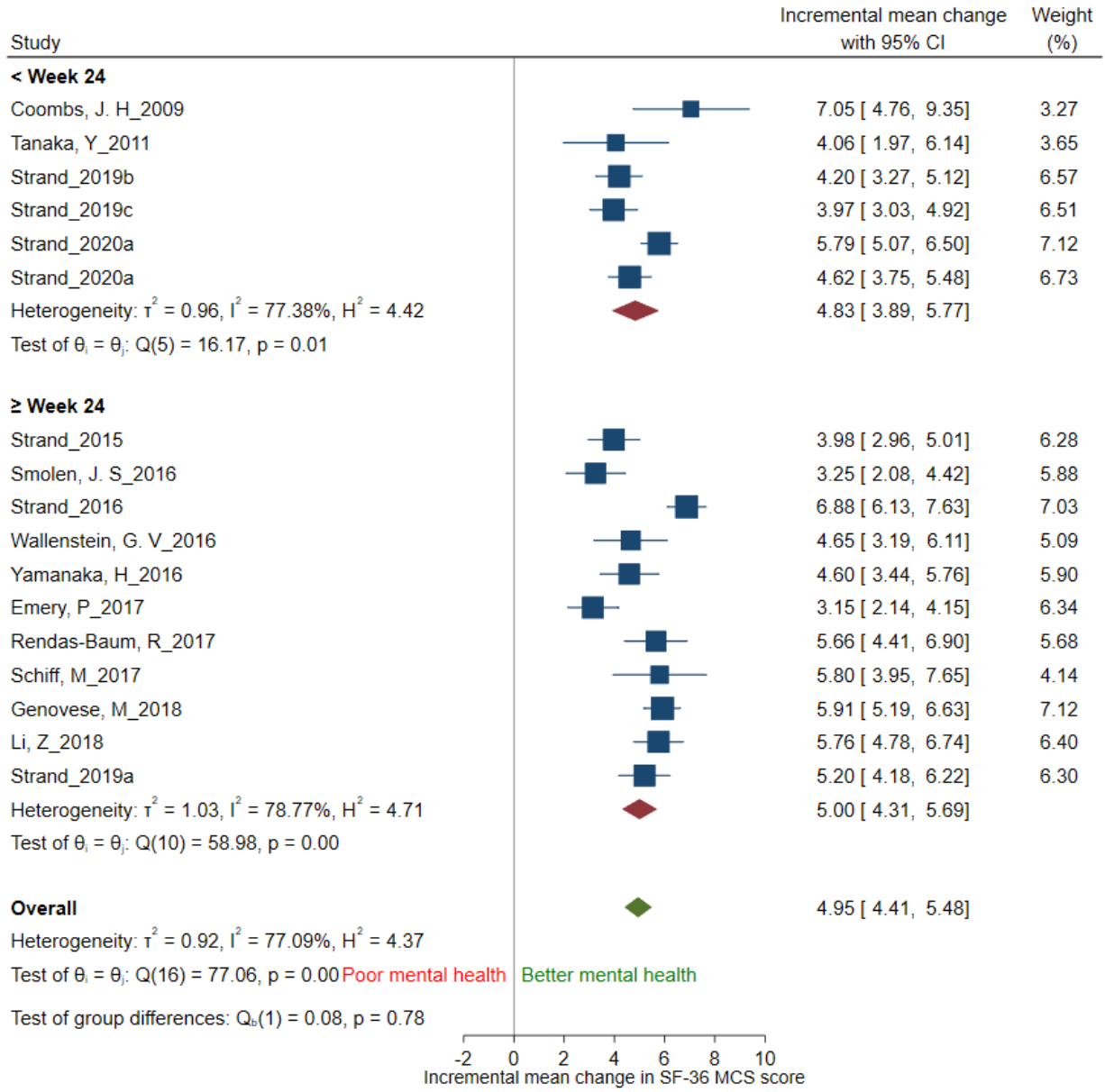
Study	Randomisation process	Deviations from the intended	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall
J.H Coobs 2009	+	+	+	+	+	+
Paul Emery 2016	+	+	+	+	+	+
Mark Genovese 2013	+	+	+	+	+	+
Edward C Keystone_2017	+	+	+	+	+	+
Zhanguo LI 2018	+	+	+	+	+	+
Regina Rendas-Baum 2015	+	+	+	+	+	+
Michael Schiff1_2017	+	+	+	+	+	+
Josef S Smolen_2016	+	+	+	+	+	+
VIBEKE STRAND 2014	+	+	+	+	+	+
Vibeke Strand1_2015	+	+	+	+	+	+
Vibeke Strand 2016	+	+	+	+	+	+
Vibeke Strand 2019	+	+	+	+	+	+
Vibeke Strand 2019	+	+	+	+	+	+
Vibeke Strand2019	+	+	+	+	+	+
Vibeke Strand 2020	+	+	+	+	+	+
V. Strand 2020	+	+	+	+	+	+
Yoshiya Tanaka 2011	+	+	+	+	+	+
G.V. Wallenstein, 2016	+	+	+	+	+	+
Hisashi Yamanaka	+	+	+	+	+	+
Mark C Genovese 2018	+	+	+	+	+	+

Supplementary figure 1 : Risk of Bias assesment of Selected Studies

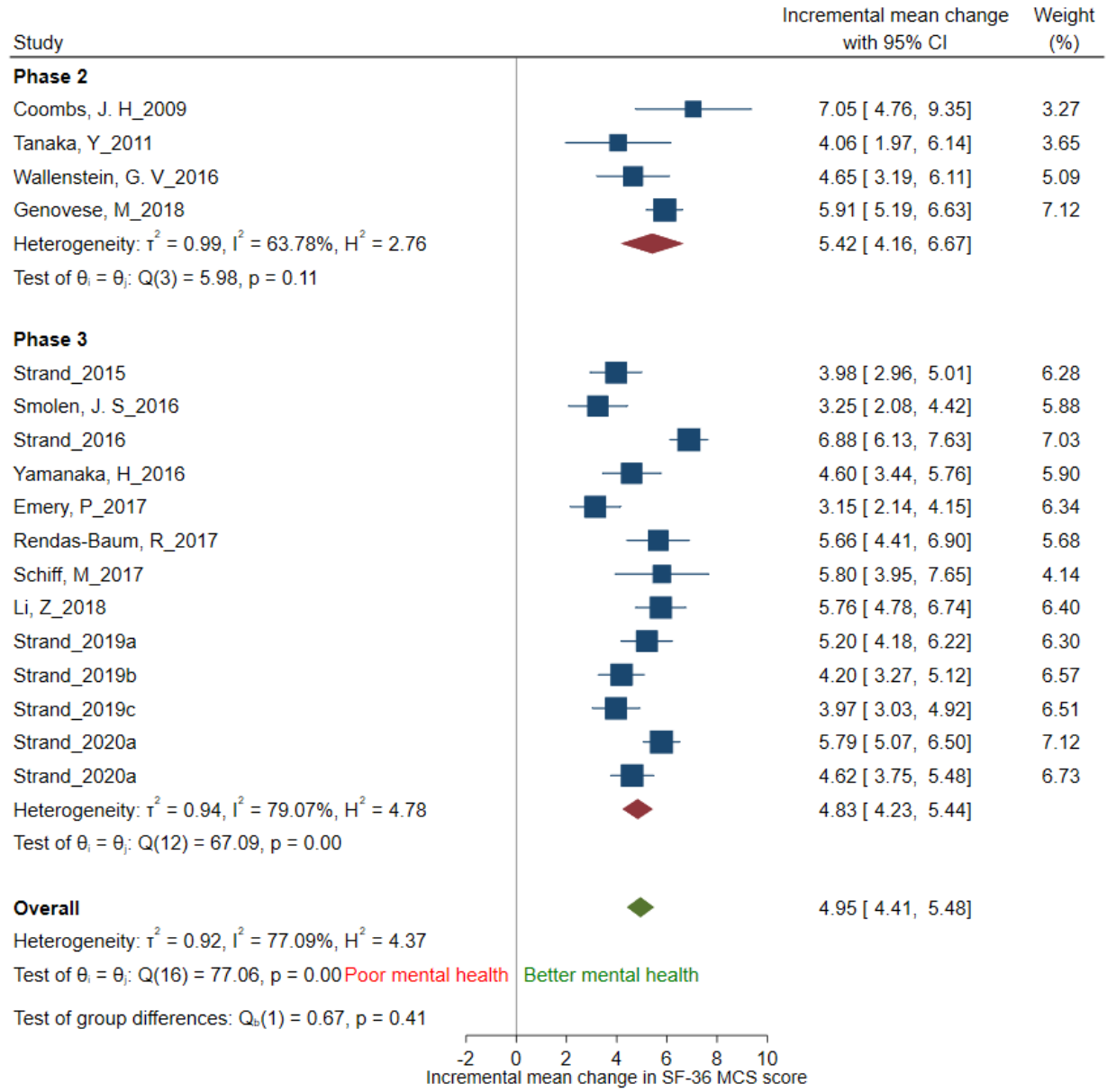
+ Low risk



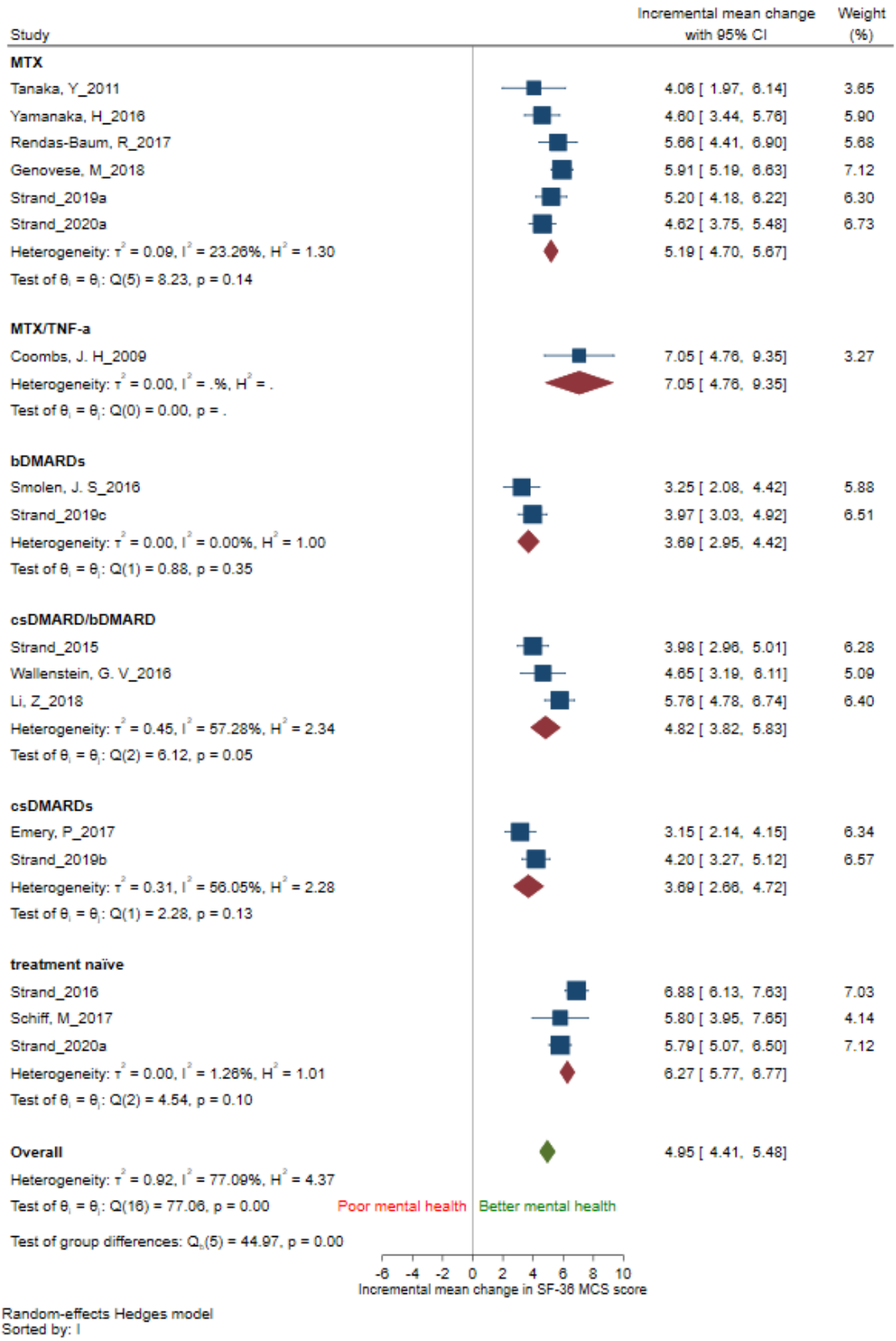
Supplementary figure 2: Funnel plot for incremental mean change of SF-36 MCS score for JAK Monotherapy



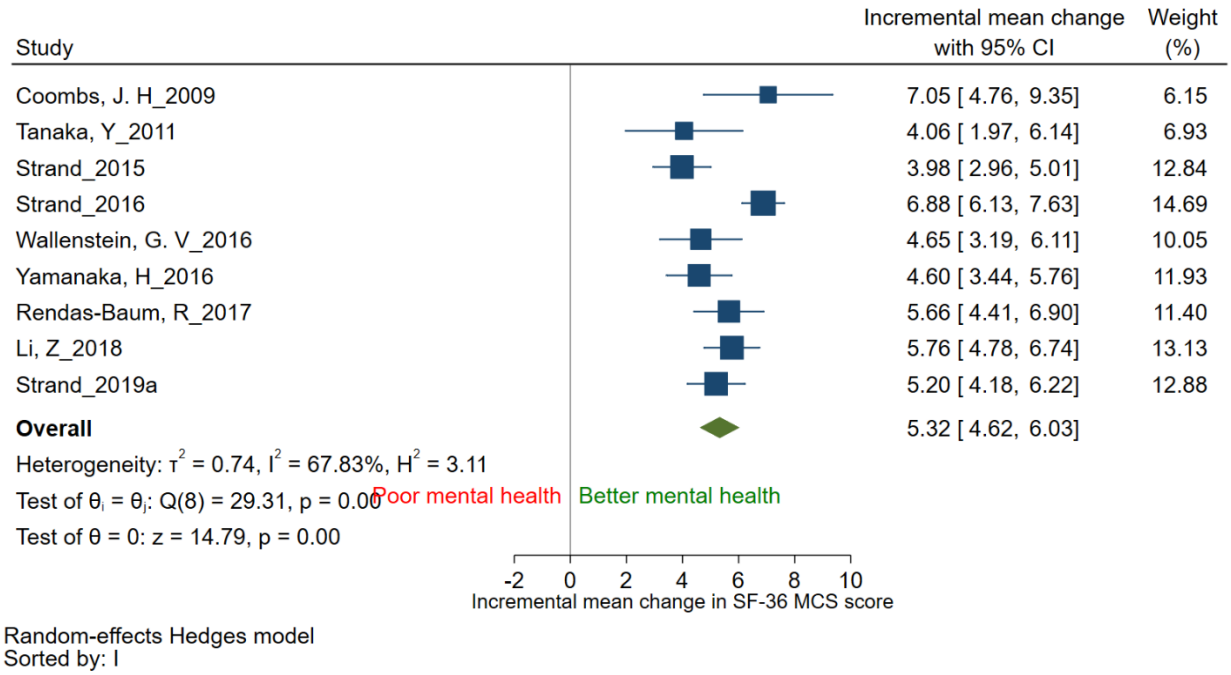
Supplementary figure 3: Sub-group analysis based on the follow-up duration



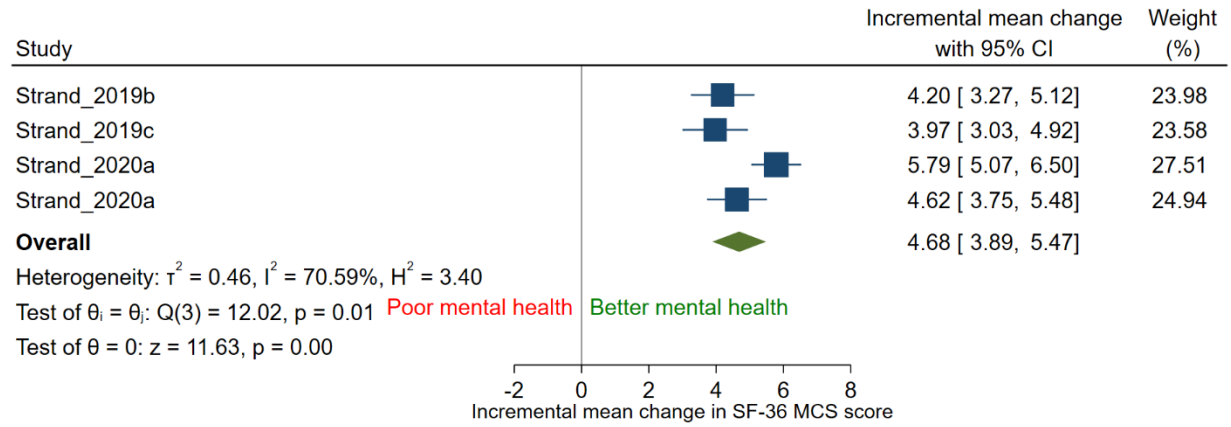
Supplementary figure 4: Sub-group analysis based on the phases of clinical trial



Supplementary figure 5: Subgroup analysis based on inadequate responses to prior treatments

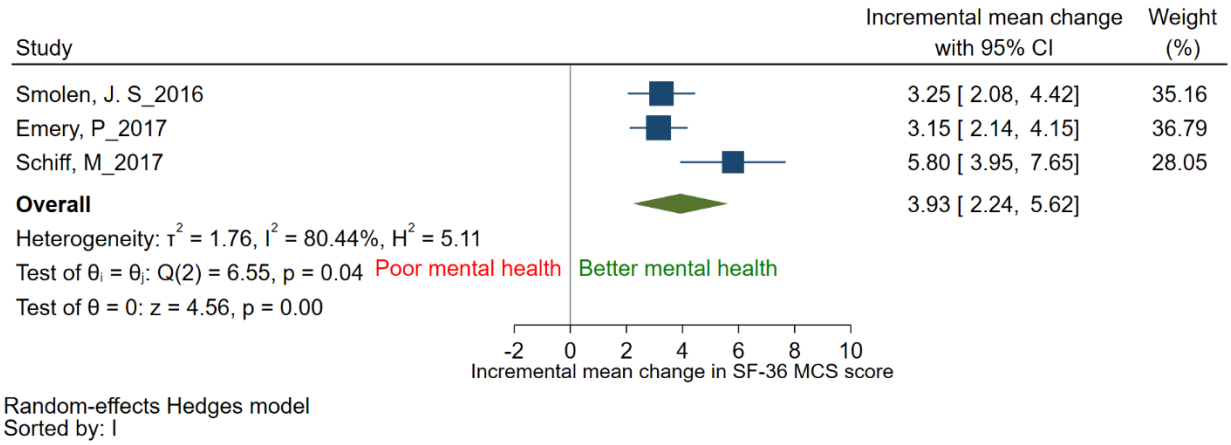


Supplementary figure 6: Separate analysis with Tofacitinib treatment

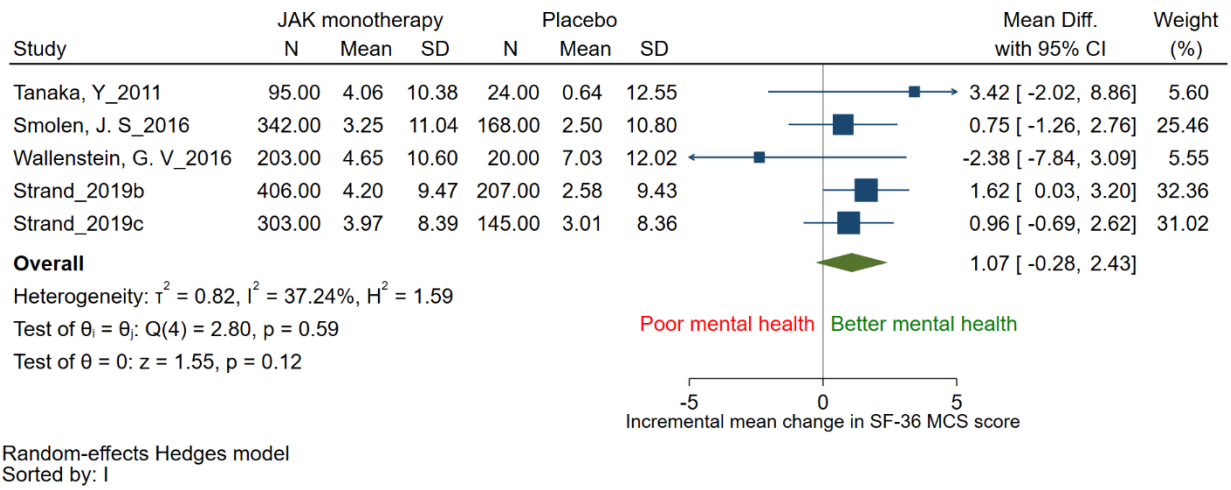


Random-effects Hedges model
Sorted by: I

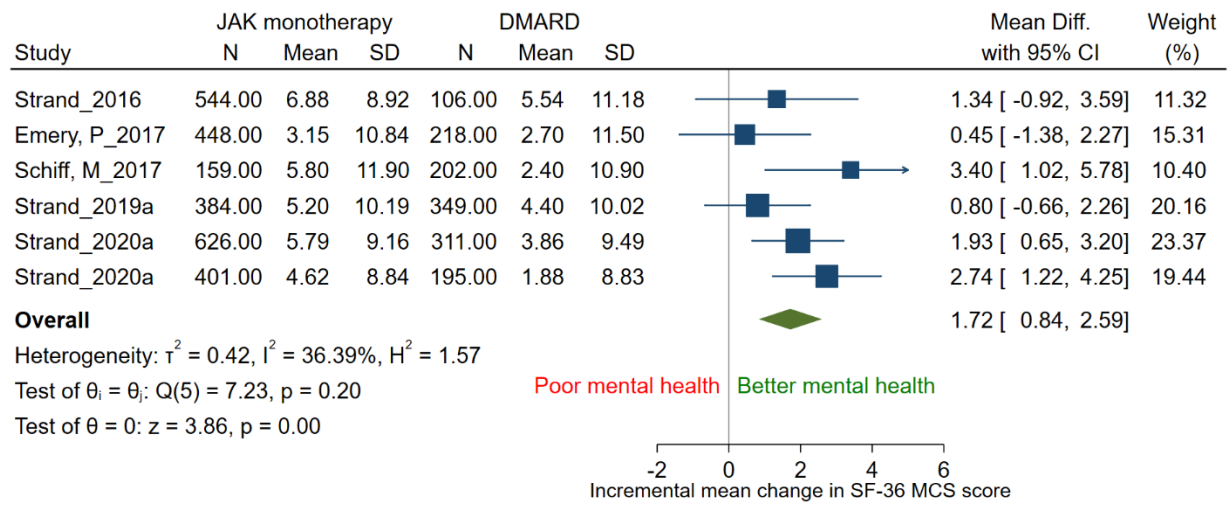
Supplementary figure 7: Separate analysis with Upadacitinib treatment



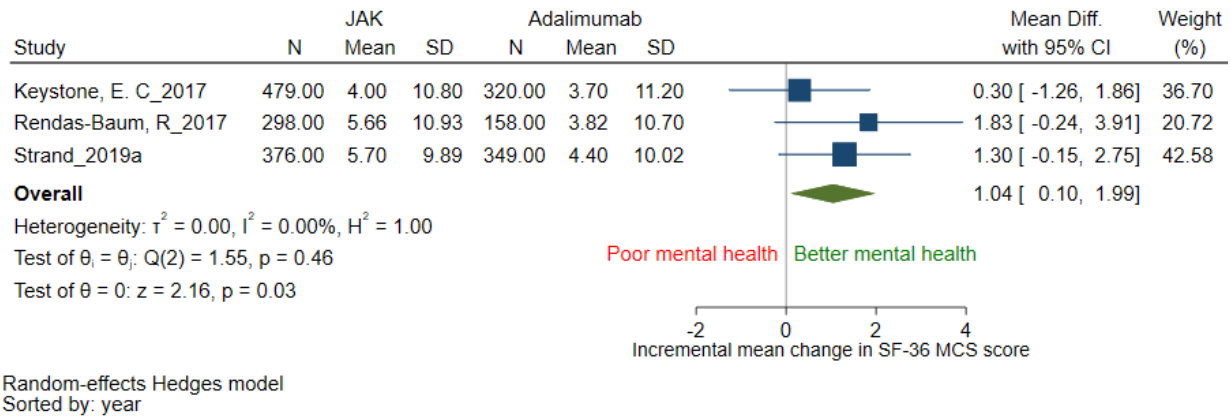
Supplementary figure 8: Sensitivity analysis with Baricitinib treatment



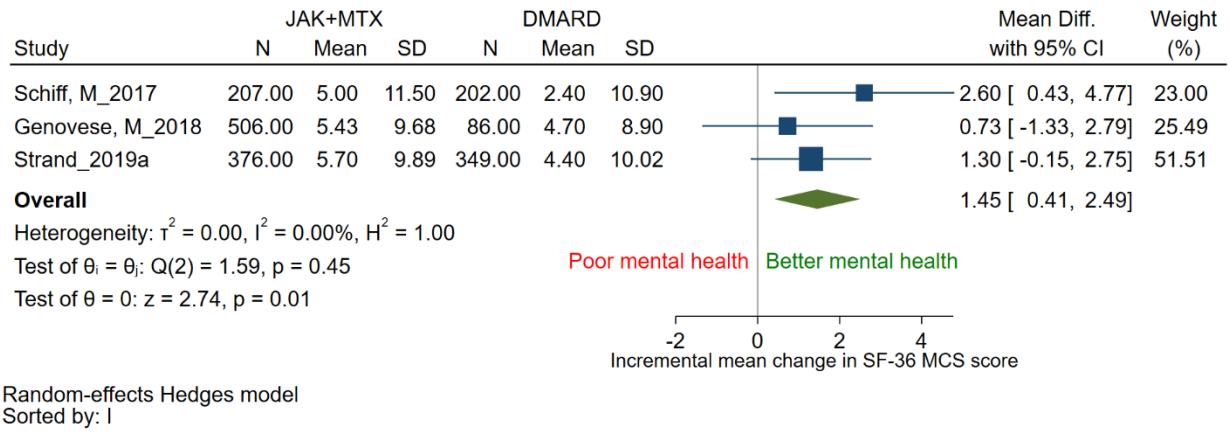
Supplementary figure 9: Sensitivity analysis of JAK Monotherapy vs Placebo



Supplementary figure 10: Sensitivity analysis of JAK Monotherapy vs DMARD

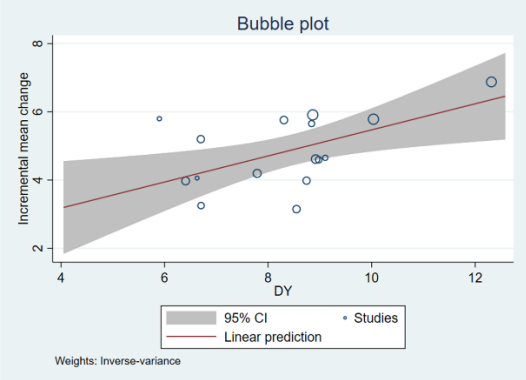


Supplementary figure 11: Sensitivity analysis of JAK vs Adalimumab

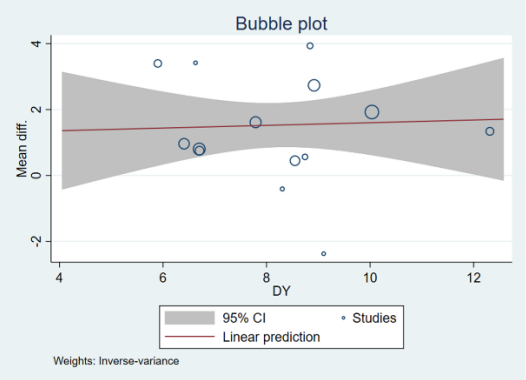


Supplementary figure 12: Sensitivity analysis of JAK+MTX vs DMARD

a.



b.



Supplementary figure 13: Metaregression with incremental mean change in SF-36 PCS

Supplementary tables

Supplementary Table 1: Search terms used in PubMed using PIOS method

PIOS	**PubMed SEARCH TERMS**	**Hits as on February 2021**
P-Population	"arthritis, rheumatoid"[MeSH Terms] OR "rheumatoid arthritis"	151,424
I-Intervention	tnf OR "Tumor Necrosis Factor" OR "JAK inhibitor" OR "JAK inhibitors" OR "Janus kinase inhibitor" OR DMARD OR "disease modifying anti-rheumatic drugs" OR biologics OR upadacitinib OR Rinvoq OR baricitinib OR Olumiant OR Simponi OR Simponi Aria OR golimumab OR certolizumab pegol OR certolizumab OR Inflectra OR infliximab-dyyb OR infliximab OR Remicade OR etanercept-szzs OR Erelzi OR etanercept OR Enbrel OR adalimumab-atto OR Amjevita OR adalimumab OR Humira OR Cyltezo OR Hyrimoz OR Cimzia OR methotrexate OR Amethopterin OR MTX OR Otrexup OR Trexall OR Rheumatrex OR Rasuvo OR tofacitinib OR Xeljanz	7,008,872
O-Outcome	"patient health questionnaire" OR "patient health questionnaire"[MeSH Terms] OR PHQ 9[Text Word] OR "General Anxiety Disorder-7" OR "GAD-7" OR DASS OR "Depression Anxiety Stress Scale" OR Anxiety OR Depression OR "HADS" OR "Hospital anxiety depression scale" OR "Hamilton Depression Rating Scale" OR "HAM-D" OR "Hamilton Anxiety Rating Scale" OR "HAM-A" OR "beck anxiety inventory" OR "beck depression inventory" OR "Center for Epidemiologic Studies Depression Scale" OR CES-D OR "Montgomery-Asberg Depression Rating Scale" OR MADRS OR "Short Form Health Survey" OR "SF-36" OR "Zung Self-Rating Depression Scale" OR "Zung Self-Rating Anxiety Scale"	696,695
S-Study design	"randomized controlled trial" OR "randomized controlled trials" OR randomized OR placebo OR randomly OR trial OR trials OR trail OR “randomised controlled trial” OR randomised OR RCT	2,404,611
PIOS		267

Supplementary Table 2: Search terms used in Scopus using PIOS method

PIOS	SCOPUS SEARCH TERMS	Hits as on February 2021
P	("Rheumatoid arthritis" OR rheumatoid)	560,278
I	tnf OR "Tumor Necrosis Factor" OR "JAK inhibitor" OR "JAK inhibitors" OR "Janus kinase inhibitor" OR DMARD OR "disease modifying anti-rheumatic drugs" OR biologics OR upadacitinib OR Rinvoq OR baricitinib OR Olumiant OR Simponi OR "Simponi Aria" OR golimumab OR "certolizumab pegol" OR certolizumab OR Inflectra OR infliximab-dyyb OR infliximab OR Remicade OR etanercept-szzs OR Erelzi OR etanercept OR Enbrel OR adalimumab-atto OR Amjevita OR adalimumab OR Humira OR Cyltezo OR Hyrimoz OR Cimzia OR methotrexate OR Amethopterin OR MTX OR Otrexup OR Trexall OR Rheumatrex OR Rasuvo OR tofacitinib OR Xeljanz	1,662,728
O	"patient health questionnaire" OR "patient health questionnaire" OR "PHQ 9" OR "General Anxiety Disorder-7" OR "GAD-7" OR DASS OR "Depression Anxiety Stress Scale" OR Anxiety OR Depression OR "HADS" OR "Hospital anxiety depression scale" OR "Hamilton Depression Rating Scale" OR "HAM-D" OR "Hamilton Anxiety Rating Scale" OR "HAM-A" OR "beck anxiety inventory" OR "beck depression inventory" OR "Center for Epidemiologic Studies Depression Scale" OR CES-D OR "Montgomery-Asberg Depression Rating Scale" OR MADRS OR "Short Form Health Survey" OR "SF-36" OR "Zung Self-Rating Depression Scale" OR "Zung Self-Rating Anxiety Scale"	7,599,618
S	"randomized controlled trial" OR "randomized controlled trials" OR randomized OR placebo OR randomly OR trial OR trials OR trail OR "randomised controlled trial" OR randomised OR RCT	7,206,915
PIOS		1998

Supplementary Table 3: Search terms used in EMBASE using PIOS method

PIOS	EMBASE SEARCH TERMS	Hits as on February 2021
P	'rheumatoid arthritis'/exp OR 'arthritis deformans' OR 'arthritis, rheumatoid' OR 'arthrosis deformans' OR 'beauvais disease' OR 'chronic articular rheumatism' OR 'chronic polyarthritis' OR 'chronic progressive poly arthritis' OR 'chronic progressive polyarthritis' OR 'chronic rheumatoid arthritis' OR 'disease, beauvais' OR 'inflammatory arthritis' OR 'polyarthritis, primary chronic' OR 'primary chronic polyarthritis' OR 'progressive polyarthritis, chronic' OR 'rheumarthritis' OR 'rheumatic arthritis' OR 'rheumatic polyarthritis' OR 'rheumatism, chronic articular' OR 'rheumatoid arthritis'	249,875
I	'tumor necrosis factor inhibitor'/exp OR 'tnf alpha inhibitor' OR 'tnf inhibitor' OR 'anti tnf agent' OR 'anti tnf alpha agent' OR 'anti tumor necrosis factor agent' OR 'anti tumour necrosis factor agent' OR 'tumor necrosis factor alpha inhibitor' OR 'tumor necrosis factor inhibitor' OR 'tumor necrosis factor inhibitors' OR 'tumour necrosis factor alpha inhibitor' OR 'tumour necrosis factor inhibitor' OR 'janus kinase inhibitor'/exp OR 'jak inhibitor' OR 'janus kinase inhibitor' OR 'janus kinase inhibitors' OR 'janus tyrosine kinase inhibitor' OR 'disease modifying antirheumatic drug'/exp OR 'disease modifying antirheumatic agent' OR 'disease modifying antirheumatic drug' OR 'disease modifying antirheumatic drugs' OR 'baricitinib'/exp OR 'baricitinib' OR 'olumiant' OR 'upadacitinib'/exp OR 'rinvoq' OR 'upadacitinib' OR 'upadacitinib 2, 3 dihydroxybutanedioate' OR 'upadacitinib hemihydrate' OR 'upadacitinib hydrate' OR 'upadacitinib tartrate' OR 'golimumab'/exp OR 'golimumab' OR 'simponi' OR 'simponi aria' OR 'certolizumab pegol'/exp OR 'certolizumab pegol' OR 'cimzia' OR 'pegylated tumor necrosis factor alpha antibody fab fragment' OR 'pegylated tumour necrosis factor alpha antibody fab fragment' OR 'certolizumab'/exp OR 'etanercept'/exp OR 'avent' OR 'benepali' OR 'brenzys' OR 'embrel' OR 'enbrel' OR 'enerceptan' OR 'erelzi' OR 'etanercept' OR 'etanercept szzs' OR 'etanercept ykro' OR 'etanercept-szzs' OR 'etanercept-ykro' OR 'eticovo' OR 'infinitam' OR 'lifmior' OR 'nepexto' OR 'opinercept' OR 'recombinant tumor necrosis factor receptor fc fusion protein' OR 'recombinant tumour necrosis factor receptor fc fusion protein' OR 'tumor necrosis factor receptor fc fusion protein' OR 'tumour necrosis factor receptor fc fusion protein' OR 'tunex' OR 'tofacitinib'/exp OR 'tasocitinib' OR 'tasocitinib citrate' OR 'tofacitinib' OR 'tofacitinib citrate' OR 'xeljanz' OR 'xeljanz xr' OR 'adalimumab'/exp OR 'ctp17' OR 'cyltezo' OR 'exemptia' OR 'gp 2017' OR 'hulio' OR 'humira' OR 'ibi303' OR 'm 923' OR 'm923' OR	283,885

	'monoclonal antibody d2e7' OR 'sb 5' OR 'sb5' OR 'amgevita' OR 'amjevita' OR 'adalimumab-bwwd' OR 'adaly' OR 'adalimumab' OR 'infliximab'/exp OR 'inflectra' OR 'infliximab' OR 'remicade' OR 'remsima' OR 'renflexis' OR 'methotrexate'/exp OR '4 amino 10 methylpteroylglutamic acid' OR '4 amino n10 methylpteroylglutamic acid' OR 'mtx' OR 'amethopterin' OR 'amethopterine' OR 'canceren' OR 'farmotrex' OR 'folex' OR 'imeth' OR 'metex' OR 'methotrexat' OR 'methotrexate' OR 'methrotrexate' OR 'methylaminopterin' OR 'meticil' OR 'metothrexate' OR 'metotrexat' OR 'metotrexate' OR 'metrex' OR 'r 9985' OR 'rheumatrex' OR 'texate' OR 'trexall' OR 'xatmep' OR 'xaken'	
O	'patient health questionnaire'/exp OR 'phq' OR 'phq-d' OR 'patient health questionnaire' OR 'personal health questionnaire' OR 'patient health questionnaire 9'/exp OR 'general anxiety disorder 7'/exp OR 'depression anxiety stress scale'/exp OR 'depression anxiety stress scale' OR 'depression, anxiety and stress scale' OR 'anxiety'/exp OR 'anxiety' OR 'depression'/exp OR 'hospital anxiety depression scale'/exp OR 'hamilton depression rating scale'/exp OR 'hamilton depression rating scale' OR 'hamilton rating scale for depression' OR 'hamilton scale' OR 'hamilton depression scale' OR 'hamilton rating scale' OR 'hamilton anxiety scale'/exp OR 'hamilton anxiety rating scale' OR 'hamilton anxiety scale' OR 'hamilton scale for anxiety' OR 'beck anxiety inventory'/exp OR 'beck anxiety inventory' OR 'beck depression inventory'/exp OR 'beck depression inventory' OR 'beck depression inventory ii' OR 'beck depression inventory revised edition' OR 'beck depression scale' OR 'montgomery asberg depression rating scale'/exp OR 'madrs' OR 'montgomery asberg depression rating' OR 'montgomery asberg depression rating scale' OR 'montgomery asberg depression scale' OR 'montgomery and asberg depression rating scale' OR madrs OR 'short form 36'/exp OR 'zung self rating depression scale'/exp OR 'zsds score' OR 'zung self rating depression scale' OR 'zung`s self-rating depression scale' OR 'zung self rating anxiety scale'/exp OR 'center for epidemiological studies depression scale'/exp OR 'ces-d' OR 'center for epidemiological studies depression scale'	815,409
S	'clinical trial'/exp OR 'clinical drug trial' OR 'clinical trial' OR 'major clinical trial' OR 'trial, clinical'	,922,856
PIOS	PIOS	652

PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE : Association between Janus kinase inhibitors therapy and mental health outcome in patients with rheumatoid arthritis: a systematic review and meta-analysis			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Yes, 1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Yes, 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	Yes, 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	Yes, 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	Yes, 6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	Yes, 5 & 6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	Yes, 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supl material
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	Yes, 6 & 7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	Yes, 6 & 7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	Yes, 6 & 7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	Yes, 7 & 8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	Yes, 7 & 8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of	Yes, 7 & 8

		consistency (e.g., I^2) for each meta-analysis.	
Section/topic	**#**	**Checklist item**	**Reported on page #**
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	Yes, 7 & 8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	Yes, 7-10
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Yes, 9 & 10, Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Yes, 9 & 10
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Yes, 9 -10 & Sup table & fig
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Yes, 9-10, Table 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Yes, 10-12 and Fig 2-4 and Supl figs
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Yes, 9 -10 & Sup table & fig
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	Yes,10-15
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	Yes, 16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	Yes, 19
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	Yes, 19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Yes, 1